

1. Analysis Overview

The biological question was whether an AI-driven bioinformatics platform could reliably replicate a validated DSP workflow on a clinically complex ECG. [Record JS00001](#), an 85-year-old male with Atrial Fibrillation (AFib) with Rapid Ventricular Response and Right Bundle Branch Block (RBBB), was selected deliberately as a stress test: its low-amplitude fibrillatory waves ($<200 \mu\text{V}$) and absent P-wave activity cause systematic Type II errors in static-threshold peak detectors, and its signal is corrupted by 50 Hz power-line interference from the Chinese hospital grid alongside respiratory wander. Any platform handling this correctly demonstrates genuine signal processing competence.

My prior independent work established a Python "white-box" pipeline using SciPy and WFDB, focused on quantifying Signal-to-Noise Ratio (SNR) improvement before clinical interpretation. I used Claude minimally for matplotlib boilerplate; it was non-contributory to filter design or statistical logic. All source code, data, and figures are available at <https://github.com/su-huang/biobuild-assessment>.

2. Developing a Workflow

I implemented a two-stage Digital Signal Processing (DSP) chain. A 4th-order Butterworth bandpass (0.5 - 45 Hz) removes DC offset and respiratory wander while preserving the QRS complex (10 - 40 Hz); zero-phase filtering via `filtfilt` prevents temporal peak shift, critical for RR-interval timing. Additionally, an IIR notch filter ($f_c = 50 \text{ Hz}$, $Q = 35$) was manually set to the Chinese power-grid standard.

Initial peak detection used a static threshold ($0.5 \times V_{\max}$), producing a greatly underestimated 48 BPM. This was a Type II error driven by the low-voltage AFib morphology (Appendix Figure 1). I corrected this by transitioning to median RR-interval BPM calculation ($\text{BPM} = 60 / \text{median}(\text{RR})$) paired with Standard Deviation of Normal-to-Normal intervals (SDNN) as a Heart Rate Variability (HRV) metric, yielding a clinically consistent $\sim 102.7 \text{ BPM}$. Welch's method Power Spectral Density (PSD) plots confirmed >3 orders of magnitude attenuation at 50 Hz, providing a quantitative audit of filter effectiveness (Appendix Figure 2).

3. Using BioBuild to Develop a Workflow

I evaluated BioBuild through two phases designed to probe the platform's range from fully autonomous inference to guided correction.

3.1 Autonomous Test

I uploaded JS00001.mat and JS00001.hef with a single objective prompt and no engineering parameters: "Perform a standard clinical diagnostic sweep. Identify the primary rhythm, detect heart rate, and visualize the most significant leads." BioBuild autonomously parsed the 16-bit ADC payload, extracted $f_s = 500 \text{ Hz}$ and gain (1000 ADU/mV), and cross-referenced SNOMED-CT codes to produce a full clinical report identifying AFib (164889003), RBBB (59118001) via rsR' morphology in Lead V1, ST-segment changes in V2 ($-74.5 \mu\text{V}$) and V5 ($+93.2 \mu\text{V}$), and a Bazett-corrected QTc of 416 ms. However, despite reporting 106.8 BPM, the plot only marked 1 R-peak in the 5-second window (Appendix Figure 3). This visualization discrepancy is detailed in Section 4.

3.2 Guided Test

Following a correction prompt specifying lead selection and filtering constraints, BioBuild refined its detection using RR Coefficient of Variation (CV = 9.7%) and SDNN (49.9 ms), correctly identifying 8 R-peaks at 111.1 BPM (Median RR: 540 ms, IQR: 95.5 - 127.7 BPM), with the frequency spectrum confirming 50 Hz notch attenuation (Appendix Figure 4).

4. BioBuild Performance Evaluation

4.1 What BioBuild Did Well

Autonomous metadata integration was the platform's strongest feature. It parsed the WFDB binary and .hea header to identify fs, ADC gain, and SNOMED-CT codes, and selected 50 Hz correctly, rather than the US default of 60 Hz for notch filtering. It used filtfilt for zero-phase filtering, translated raw μV -level features into clinical outputs (QTc, SDNN, ST quantification), and self-corrected when guided.

4.2 Hallucinations and Incorrect Assumptions

The autonomous run produced a visual-calculation discrepancy, where it reported 106.8 BPM while only 1 R-peak was plotted. One peak in 5 seconds implies ~ 12 BPM, indicating the numerical backend and plotting engine used different internal thresholds. This is a silent failure that could go undetected by a non-expert user. Additionally, the initial output included a Heart Rate SD of ± 197.7 BPM, 1.8x the reported mean, a statistically degenerate result that BioBuild did not flag as low-confidence.

4.3 Questionable Methods, Tool Choices, and Parameter Settings

Peak-detection parameters (prominence, distance, height) were never exposed, making the initial failure of ~ 8 of 9 peaks unauditable. PSD plots were also absent from the default output, preventing spectral verification of the notch filter without an explicit follow-up prompt.

4.4 Reliability for Real Research Use

Verdict: Conditionally Reliable

BioBuild is a powerful accelerator for exploratory prototyping and clinical hypothesis generation, but is not suitable as a standalone validator for peer-reviewed research. The visualization-calculation discrepancy is a latent failure mode dangerous in non-expert hands. It functions best as a diagnostic co-pilot paired with independent signal verification.

4.5 Recommended Platform Improvements

BioBuild could benefit from four targeted improvements. First, automated sanity checks should flag discrepancies between the visualized peak count and reported BPM exceeding $\sim 10\%$. This would have surfaced the ~ 95 BPM desynchronization in the autonomous run. Second, peak-detection parameters (prominence, distance, height) should be exposed so researchers can audit and reproduce results, which is a baseline requirement for any research-grade workflow. Third, PSD plots should be included by default in all filtering outputs, providing inherent spectral verification of notch and bandpass effectiveness. Finally, when power-line interference is detected, the platform should explicitly confirm the regional frequency standard (50 Hz vs. 60 Hz) with the user rather than inferring it silently. This is a particularly important safeguard for international datasets where an incorrect assumption directly corrupts the filtered signal.

Appendix: Figures 1 - 4

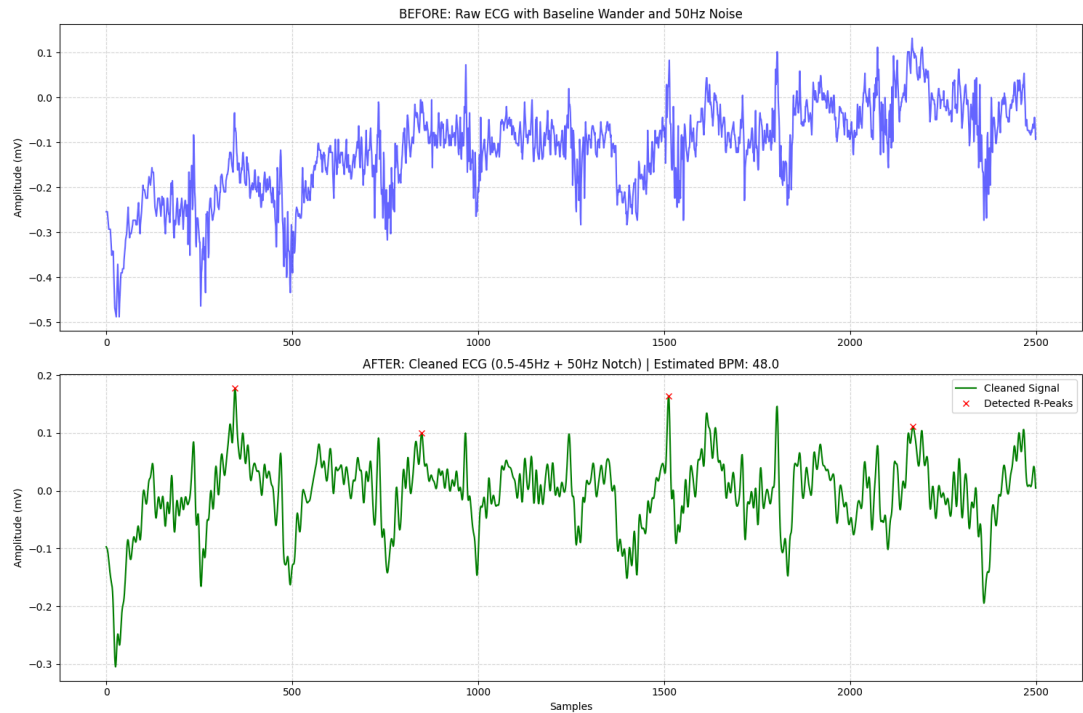


Figure 1: Raw vs. filtered Lead I (JS00001, first 5 s). Static threshold captures only 4 R-peaks, yielding an erroneous 48 BPM.

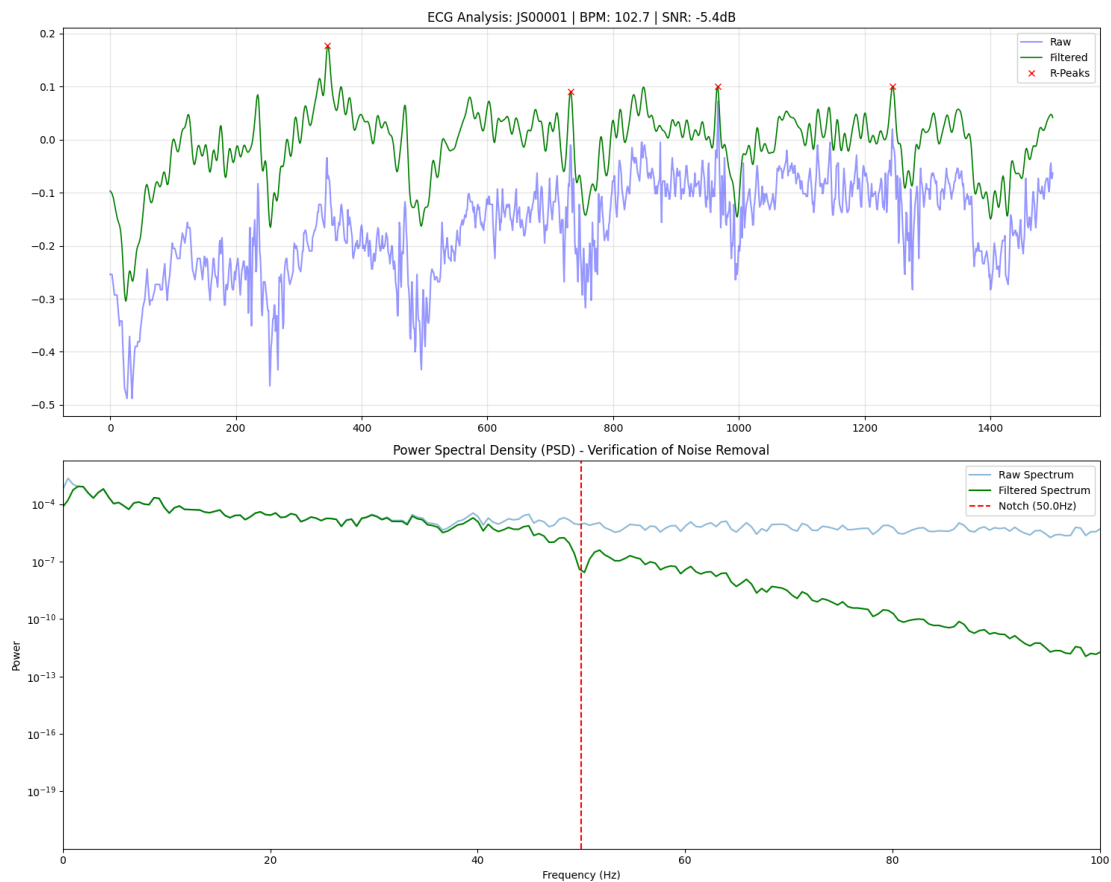


Figure 2: Refined pipeline output. PSD confirms 50 Hz attenuation; median RR method identifies 4 peaks at 102.7 BPM.



Figure 3: BioBuild autonomous output. 106.8 BPM reported but only 1 R-peak marked in 5 s, implying ~12 BPM).

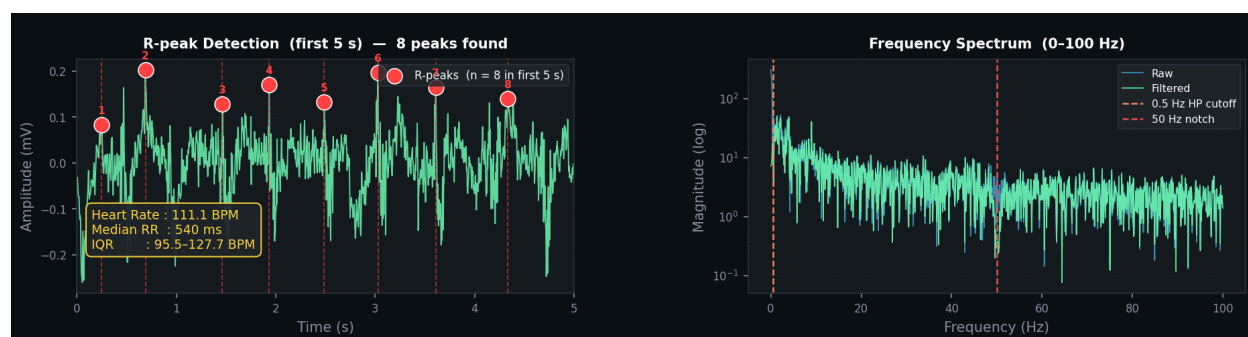


Figure 4: BioBuild corrected output. 8 R-peaks identified at 111.1 BPM and frequency spectrum confirms 50 Hz notch.